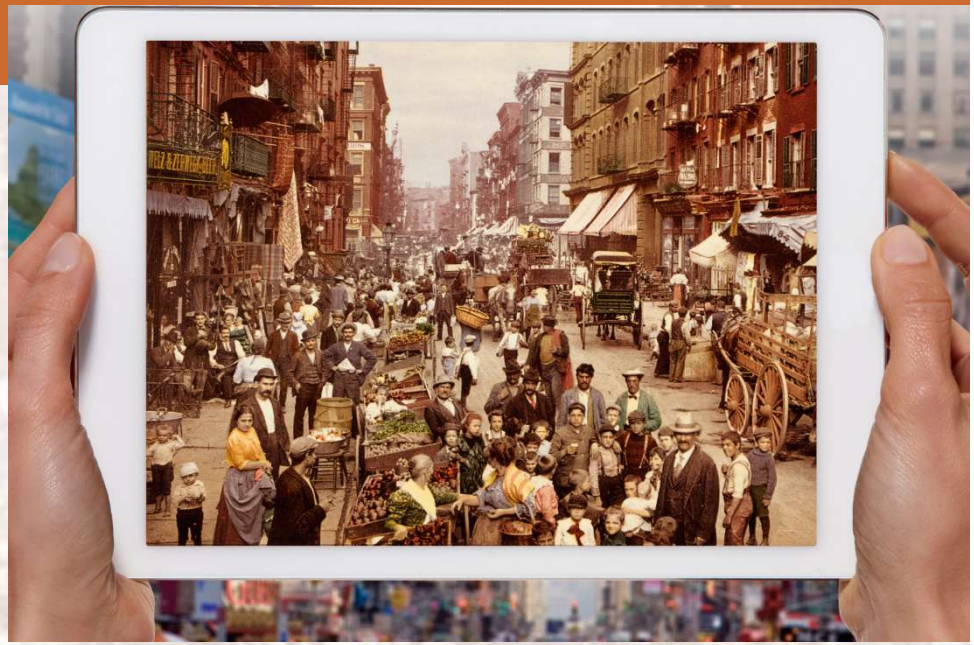
# 經濟學



CHAPTER 3

# Interdependence and the Gains from Trade

National Central University

# Look for the answers to these questions:

- Why do people – and nations – choose to be economically interdependent?
- How can trade make everyone better off?
- What is absolute advantage?
- What is comparative advantage?
- How are these concepts similar?
- How are they different?

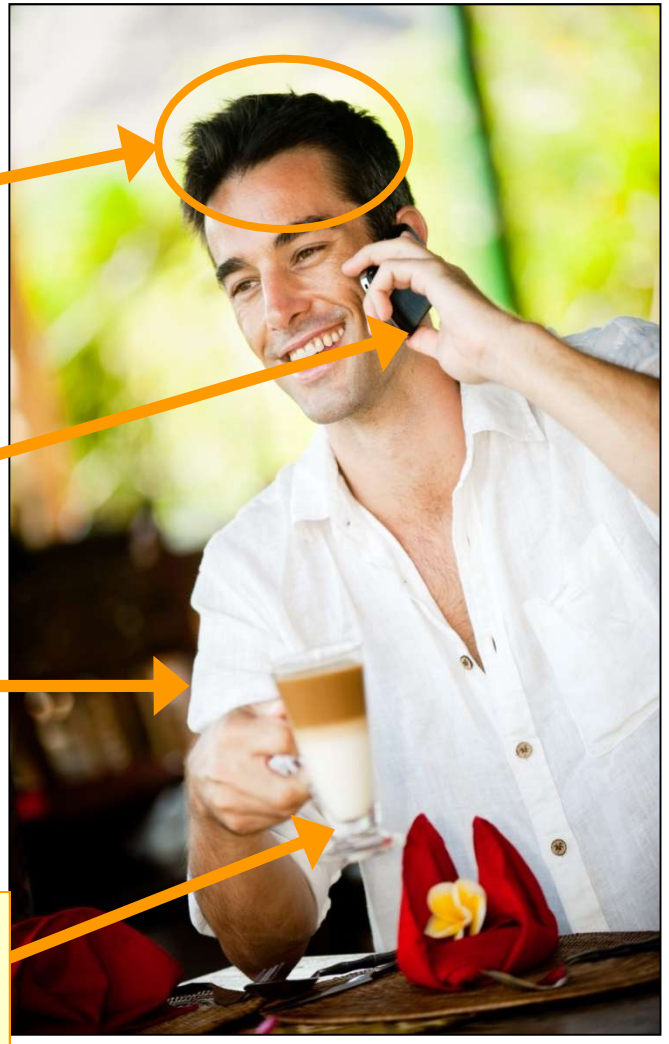
# Review Question

- Maria can read 20 pages of economics in an hour.
- She can also read 50 pages of sociology in an hour
- She spends 5 hours per day studying
- What is Maria's opportunity cost of reading 100 pages of sociology?



# Interdependence

Every day you rely on many people from around the world, most of whom you've never met, to provide you with the goods and services you enjoy.





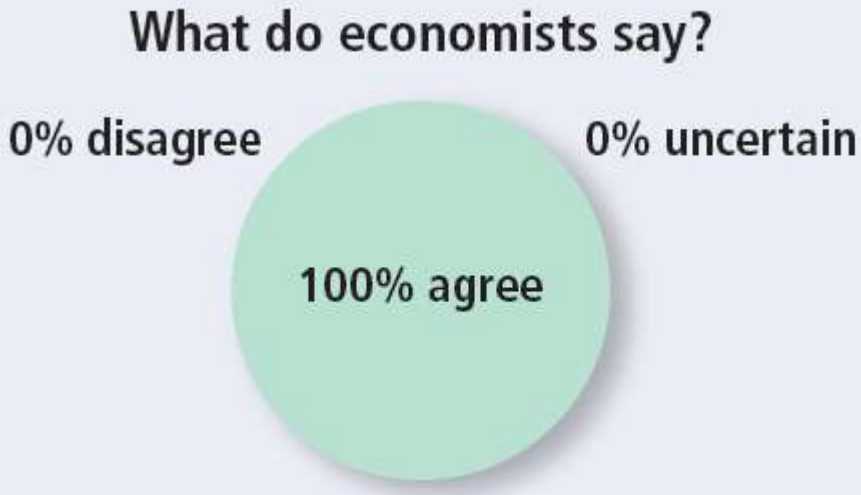
# Interdependence

- "Trade can make everyone better off"
  - One of the Ten Principles from Chapter 1
  - We now learn why people – and nations – choose to be interdependent
  - And how they can gain from trade

# ASK THE EXPERTS

## Trade between China and the United States

*"Trade with China makes most Americans better off because, among other advantages, they can buy goods that are made or assembled more cheaply in China."*





# Our Example

- Two countries:
  - The U.S. and Japan
- Two goods:
  - Computers and wheat
- One resource:
  - Labor, measured in hours
- How much of both goods each country produces and consumes
  - If the country chooses to be self-sufficient
  - If it trades with the other country

兩種選擇

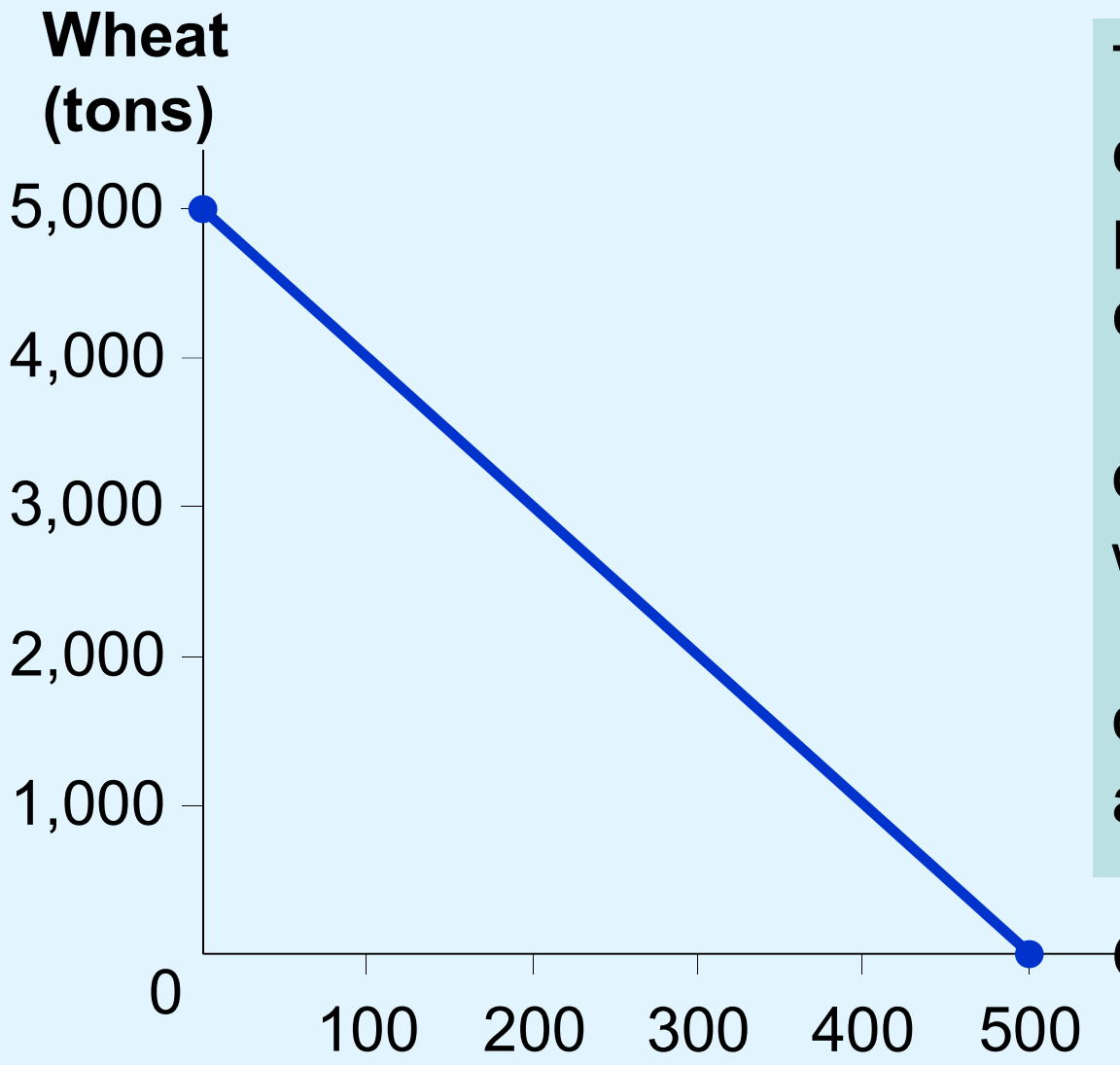


# Our Example

- Production Possibilities in the U.S.
  - The U.S. has 50,000 hours of labor available for production, per month
  - Producing one computer requires 100 hours of labor
  - Producing one ton of wheat requires 10 hours of labor

# The U.S. PPF

生產可能線 Production possibilities frontier (PPF)

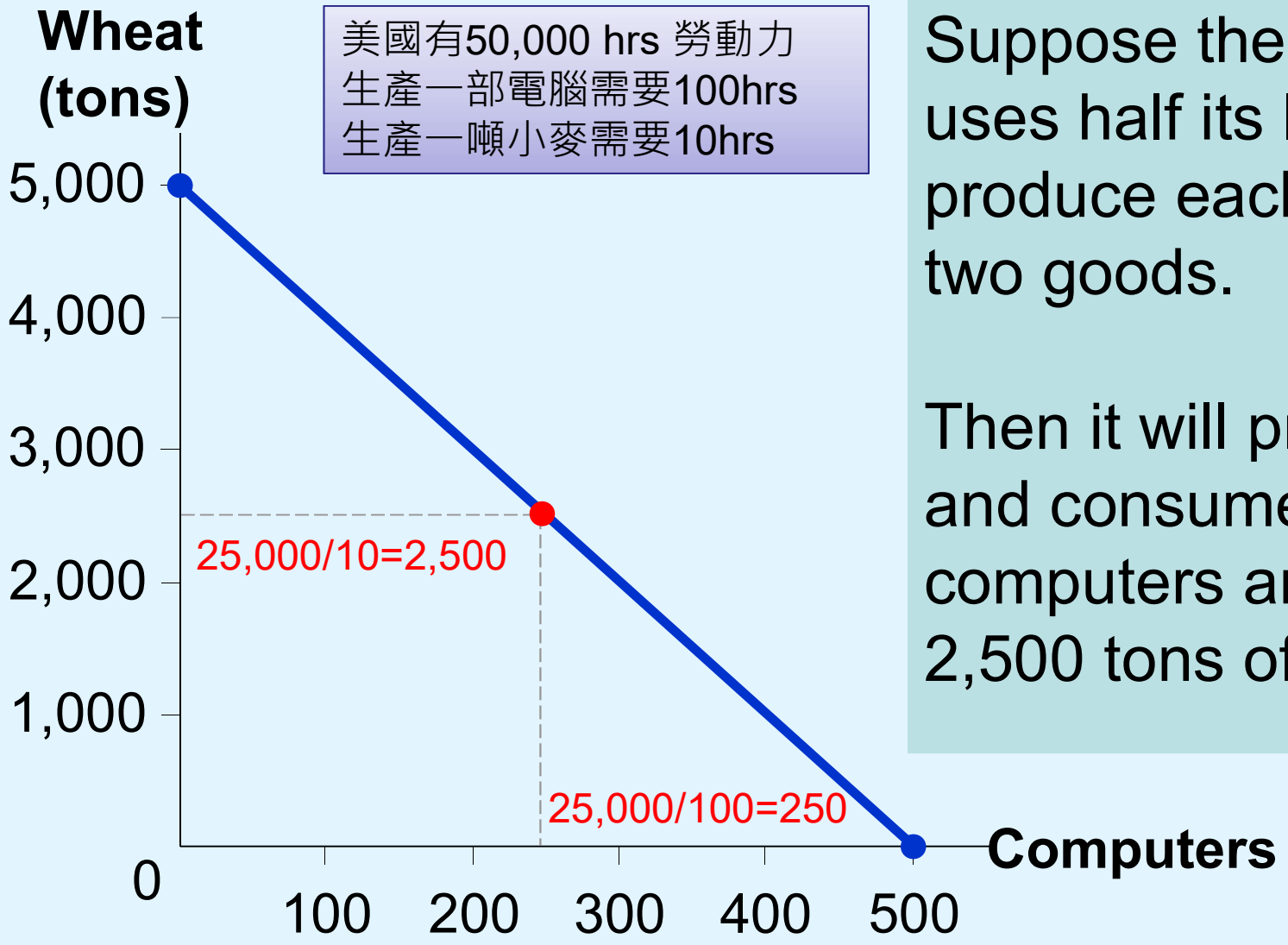



The U.S. has enough labor to produce 500 computers,

or 5,000 tons of wheat,

or any combination along the PPF.

# The U.S. Without Trade




美國有50,000 hrs 勞動力
生產一部電腦需要100hrs
生產一噸小麥需要10hrs

Suppose the U.S. uses half its labor to produce each of the two goods.

Then it will produce and consume 250 computers and 2,500 tons of wheat.

要決定PPF上的哪一點取決於社會偏好(the preferences of society)

# Active Learning 1 Derive Japan's PPF

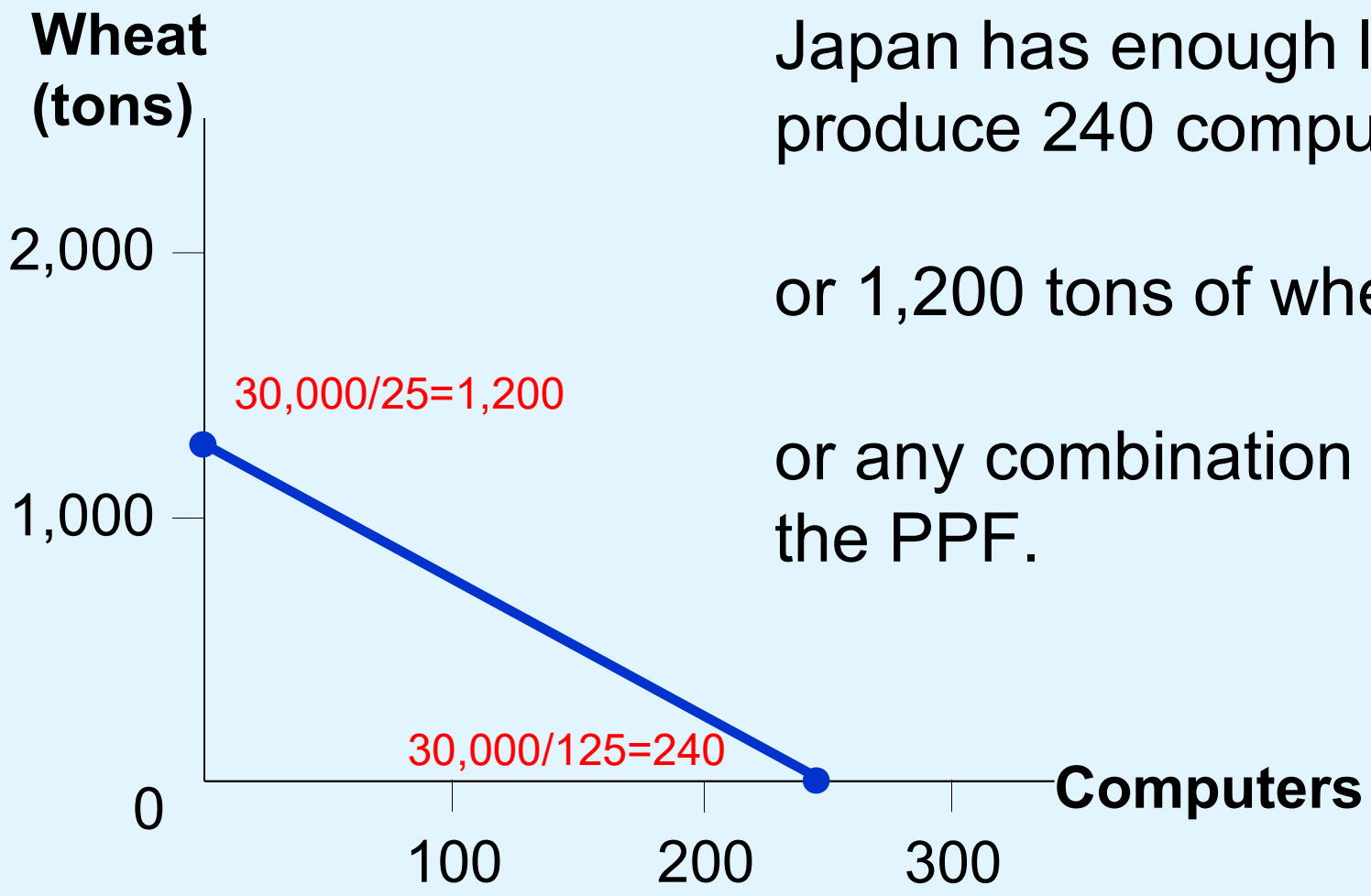
Use the following information to draw Japan's PPF.

- Japan has 30,000 hours of labor available for production, per month.
- Producing one computer requires 125 hours of labor.
- Producing one ton of wheat requires 25 hours of labor.

- Your graph should measure computers on the horizontal axis.

# Active Learning 1

## Japan's PPF



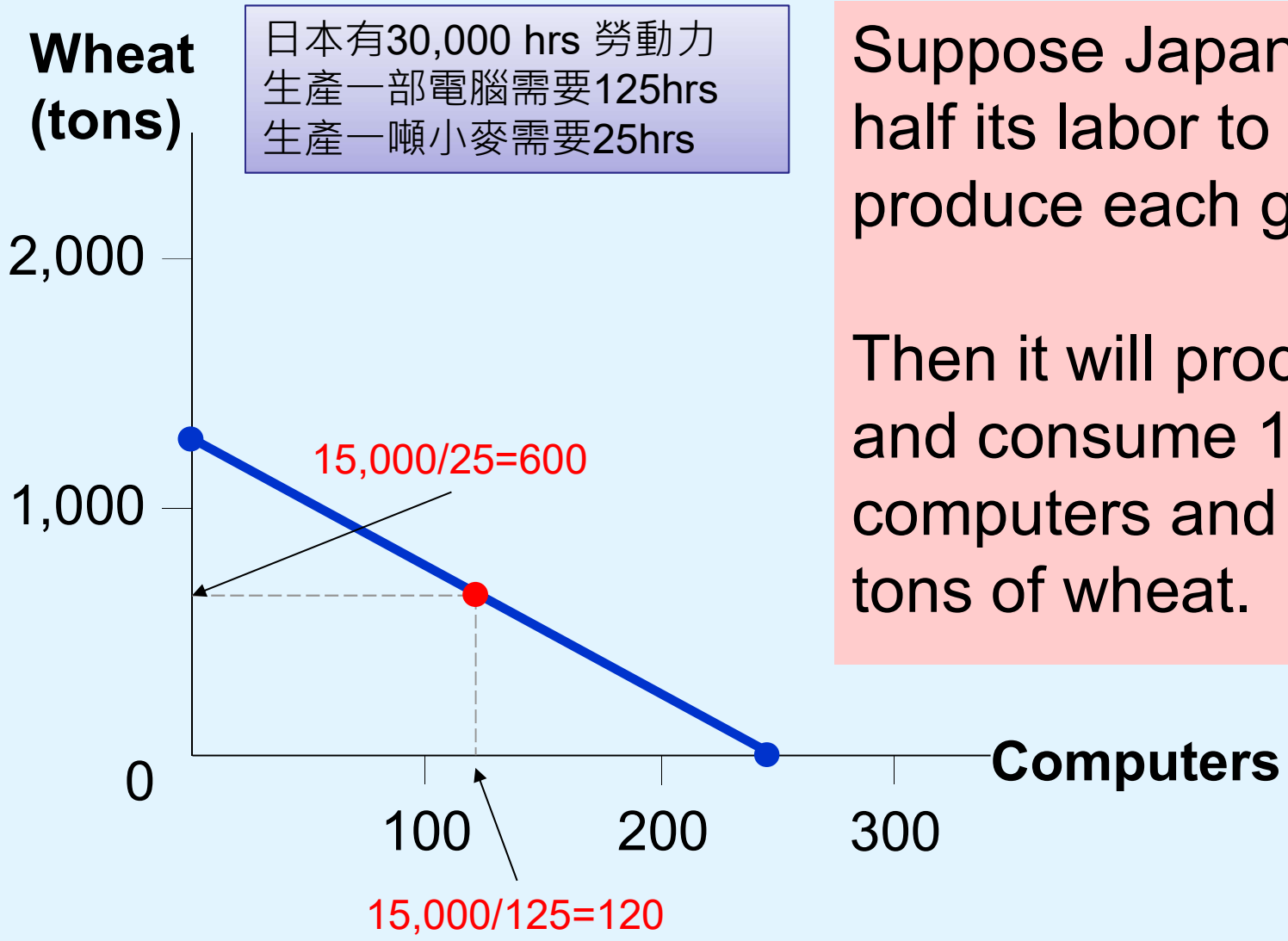

Japan has enough labor to produce 240 computers,

or 1,200 tons of wheat,

or any combination along the PPF.

日本有30,000 hrs 勞動力
生產一部電腦需要125hrs
生產一噸小麥需要25hrs

# Japan Without Trade




Suppose Japan uses half its labor to produce each good.

Then it will produce and consume 120 computers and 600 tons of wheat.



# Consumption With and Without Trade

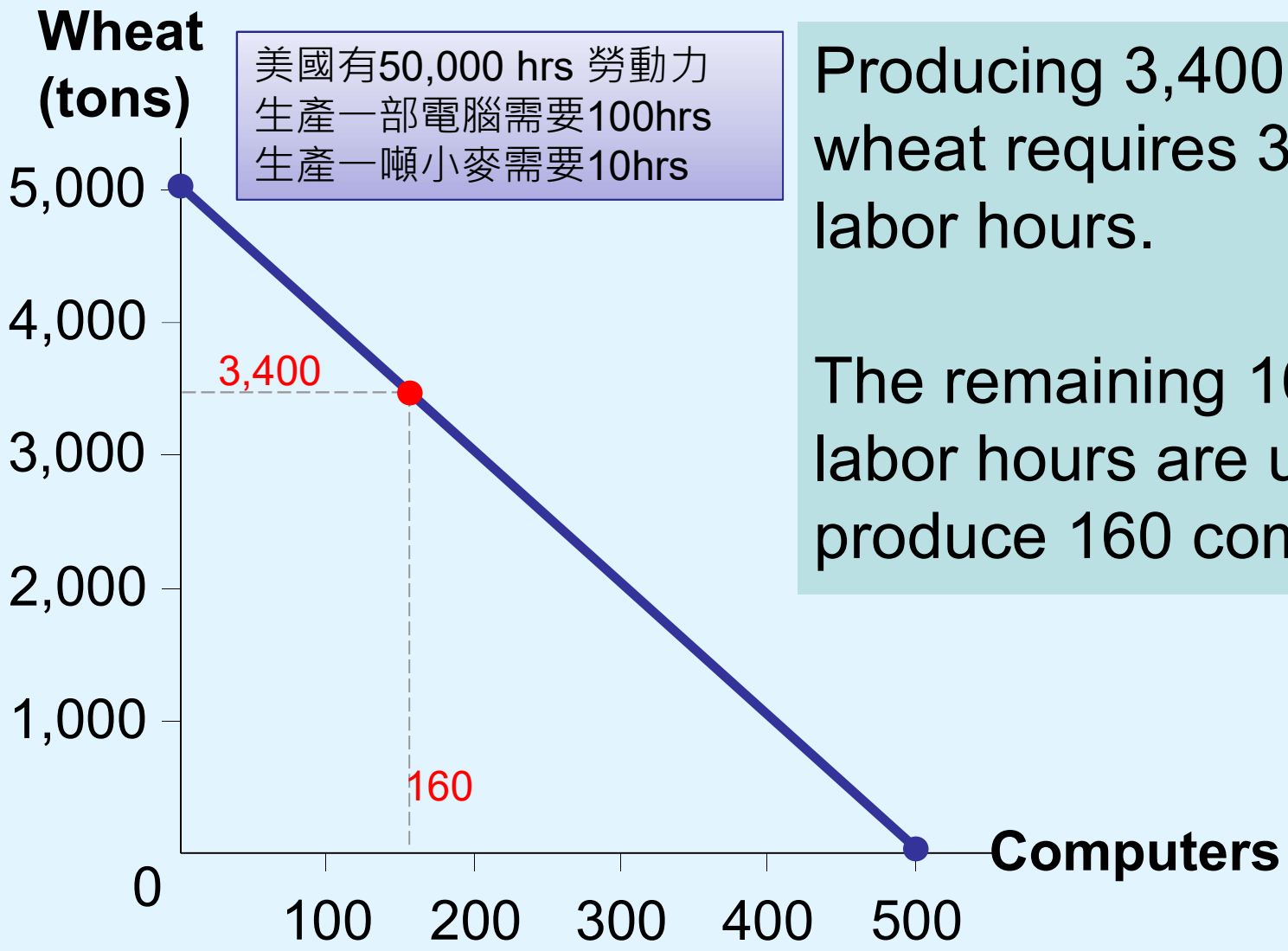
- Without trade:
  - U.S. consumers get 250 computers and 2500 tons wheat
  - Japanese consumers get 120 computers and 600 tons wheat
- Comparison
  - Consumption without trade vs. consumption with trade
  - We need to see how much of each good is produced and traded by the two countries

## Active Learning 2 — Production under trade

A. Suppose the U.S. produces 3400 tons of wheat.
   - How many computers would the U.S. be able to produce with its remaining labor?
   - Draw the point representing this combination of computers and wheat on the U.S. PPF.

B. Suppose Japan produces 240 computers.
   - How many tons of wheat would Japan be able to produce with its remaining labor?
   - Draw this point on Japan's PPF.

# Active Learning 2 A. U.S. Production With Trade




美國有50,000 hrs 勞動力
生產一部電腦需要100hrs
生產一噸小麥需要10hrs

Producing 3,400 tons of wheat requires 34,000 labor hours.

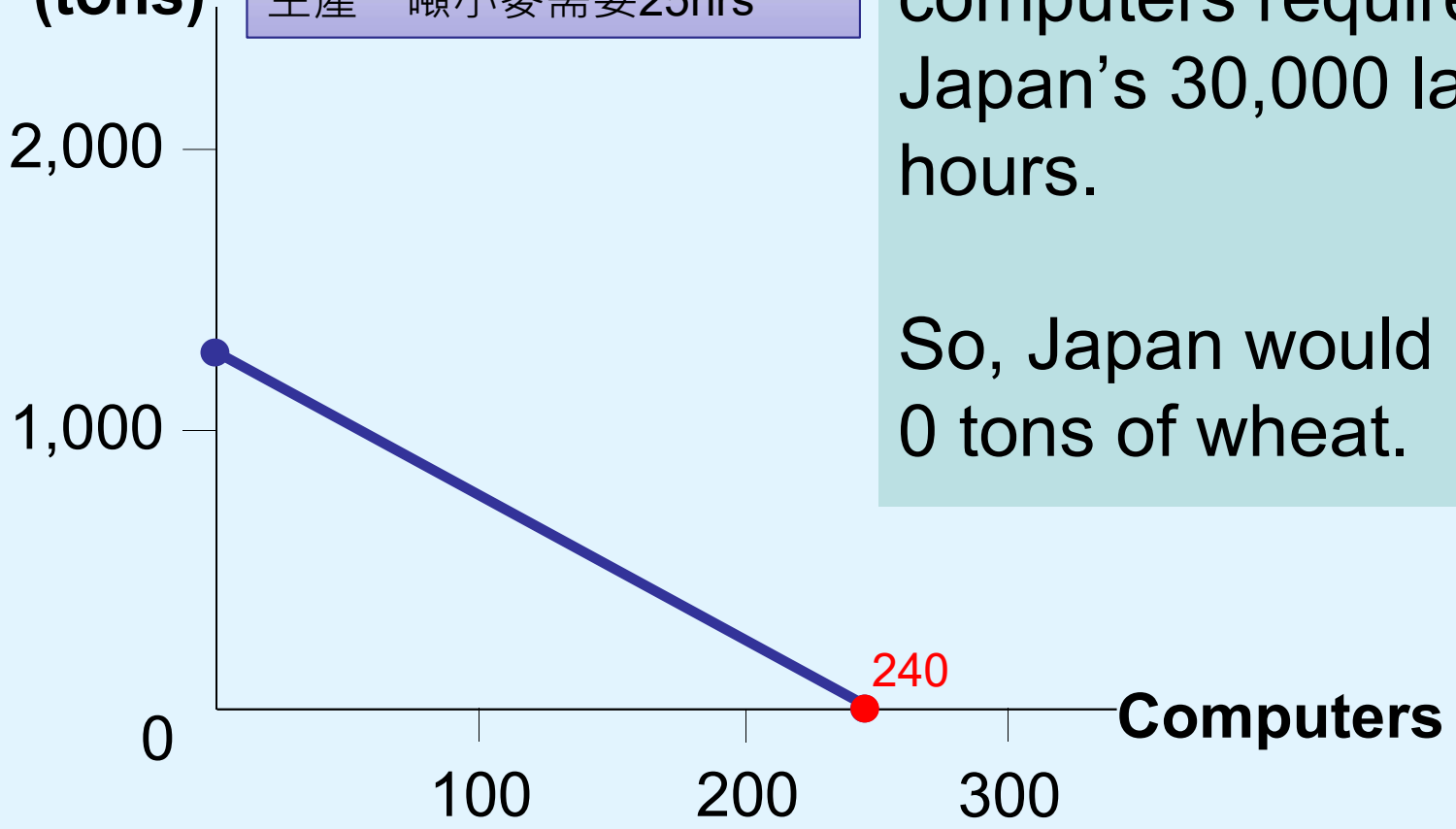
The remaining 16,000 labor hours are used to produce 160 computers.

小麥工時=3,400x10=34,000 hrs
剩餘工時=50,000-34,000=16,000 hrs

電腦產能=16,000/100=160 台

# Active Learning 2 **B.** Japan's Production With Trade

日本有30,000 hrs 勞動力
生產一部電腦需要125hrs
生產一噸小麥需要25hrs




Producing 240 computers requires all of Japan's 30,000 labor hours.

So, Japan would produce 0 tons of wheat.

電腦工時=240x125=30,000 hrs
剩餘工時=30,000-30,000=0 hrs

小麥產能=0/25=0 頓



# Exports and Imports

- Imports
  - Goods produced abroad and sold domestically
- Exports
  - Goods produced domestically and sold abroad

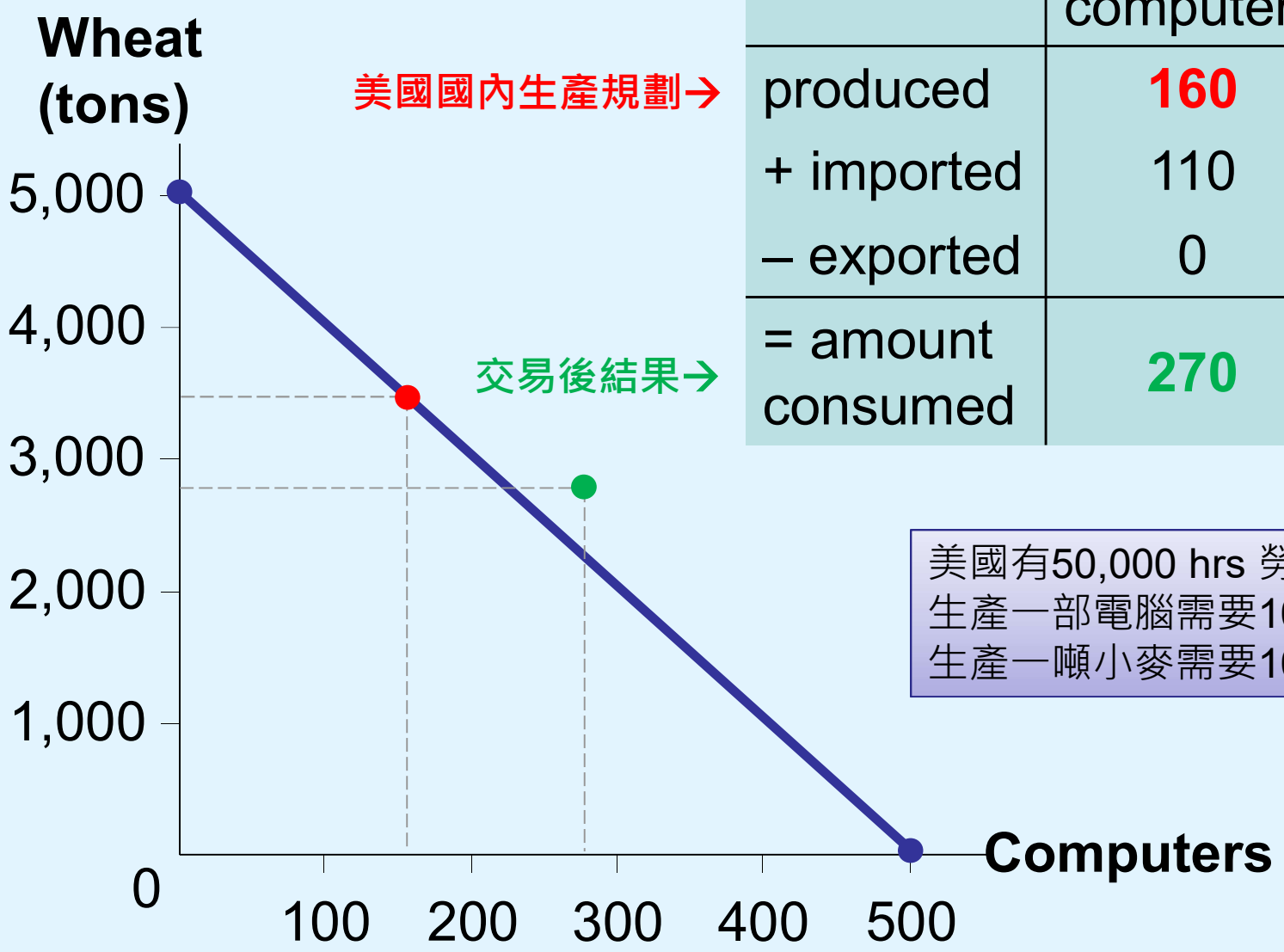
## Active Learning 3 Consumption under trade

Suppose the U.S. exports 700 tons of wheat to Japan, and imports 110 computers from Japan. (Japan imports 700 tons wheat and exports 110 computers.)

A. How much of each good is consumed in the U.S.? Plot this combination on the U.S. PPF.

B. How much of each good is consumed in Japan? Plot this combination on Japan's PPF.

# Active Learning 3 **A.** U.S. Consumption With Trade




美國國內生產規劃→

交易後結果→

| | computers | wheat |
|---|---|---|
| produced | **160** | **3400** |
| + imported | 110 | 0 |
| – exported | 0 | 700 |
| = amount consumed | **270** | **2700** |

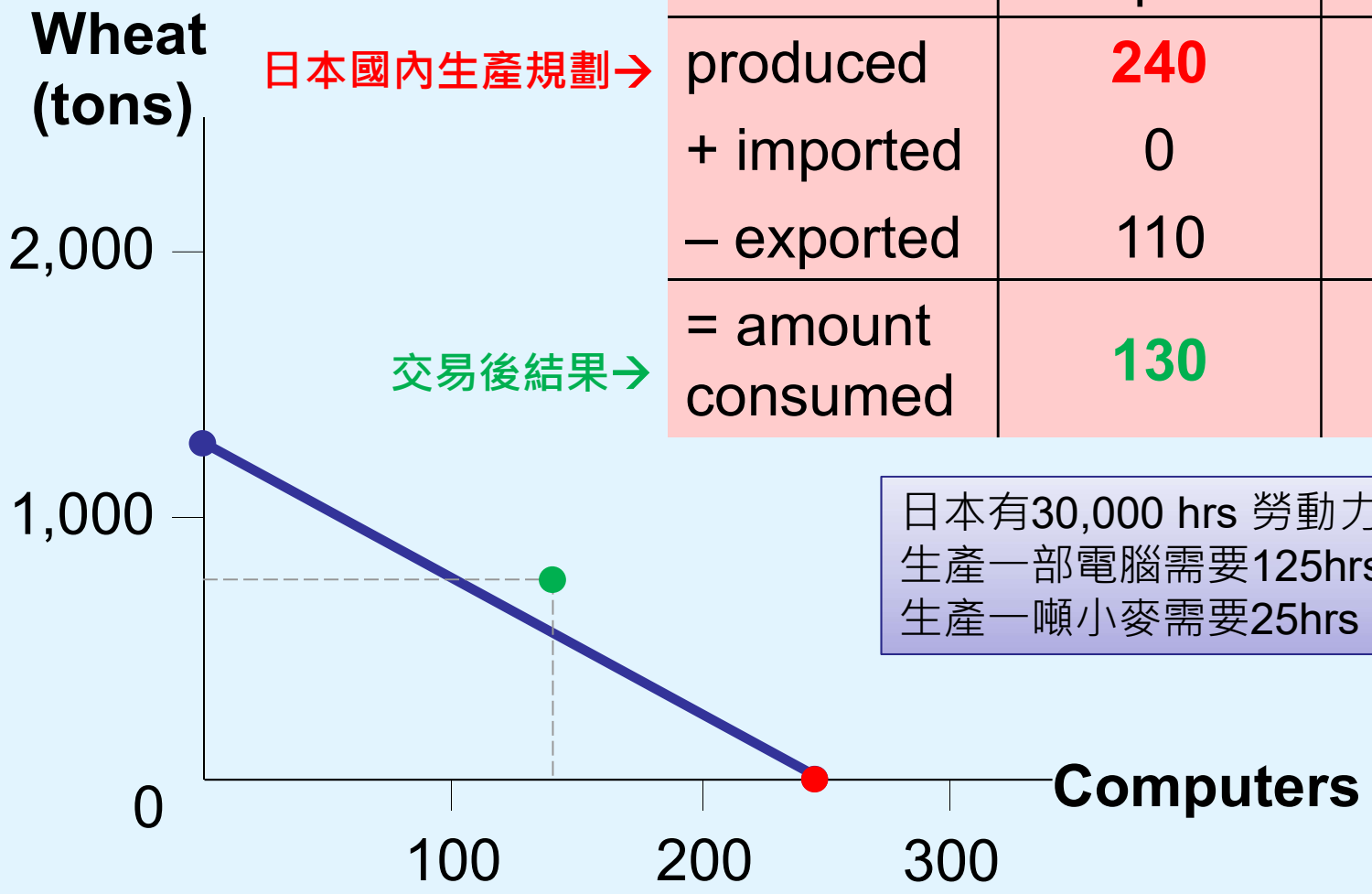
美國有50,000 hrs 勞動力
生產一部電腦需要100hrs
生產一噸小麥需要10hrs

# Active Learning 3 B. Japan's Consumption With Trade

| | computers | wheat |
|---|---|---|
| produced | **240** | **0** |
| + imported | 0 | 700 |
| – exported | 110 | 0 |
| = amount consumed | **130** | **700** |

日本國內生產規劃→

交易後結果→




日本有30,000 hrs 勞動力
生產一部電腦需要125hrs
生產一噸小麥需要25hrs

# Trade Makes Both Countries Better Off

美國有50,000 hrs 勞動力
生產一部電腦需要100hrs
生產一噸小麥需要10hrs

| U.S. | | | |
|---|---|---|---|
| | consumption without trade | consumption with trade | gains from trade |
| computers | 250 (50%) | 270 (160+110) | 20 |
| wheat | 2500 (50%) | 2700 (3,400-700) | 200 |

日本有30,000 hrs 勞動力
生產一部電腦需要125hrs
生產一噸小麥需要25hrs

| Japan | | | |
|---|---|---|---|
| | consumption without trade | consumption with trade | gains from trade |
| computers | 120 (50%) | 130 (240-110) | 10 |
| wheat | 600 (50%) | 700 (0+700) | 100 |



# Where Do These Gains Come From?

- Absolute advantage:
  - The ability to produce a good using fewer inputs than another producer
  - The U.S. has absolute advantage in wheat
    - Producing a ton of wheat uses 10 labor hours in the U.S. vs. 25 in Japan
  - The U.S. has absolute advantage in computers
    - Producing one computer requires 125 labor hours in Japan, but only 100 in the U.S.



# Where Do These Gains Come From?

The U.S. has an absolute advantage in both goods!

- So why does Japan specialize in computers?
- Why do both countries gain from trade?

- Two countries can gain from trade
  - When each specializes in the good it produces at lowest opportunity cost



# 補充說明

- 根據亞當斯密(Adam Smith)學說,一個國家應該生產具絕對優勢的產品,並透過交易換來不具絕對優勢的產品
- 但亞當斯密學說無法解釋先進國家與落後國家的交易行為
- 李嘉圖(David Ricardo)提出”比較優勢”理論來解釋此現象



# Two Measures of the Cost of a Good

- Absolute advantage
  - Measures the cost of a good in terms of the inputs required to produce it

美國有50,000 hrs 勞動力
生產一部電腦需要100hrs
生產一噸小麥需要10hrs

日本有30,000 hrs 勞動力
生產一部電腦需要125hrs
生產一噸小麥需要25hrs

- Another measure of cost: opportunity cost
  - The opportunity cost of a computer = amount of wheat that could be produced using the labor needed to produce one computer

美國有50,000 hrs 勞動力
生產一部電腦的機會成本為10噸小麥
生產一噸小麥的機會成本為1/10台電腦

日本有30,000 hrs 勞動力
生產一部電腦的機會成本為5噸小麥
生產一噸小麥的機會成本為1/5台電腦



# Comparative Advantage

- Comparative advantage
  - The ability to produce a good at a lower opportunity cost than another producer
- Principle of comparative advantage
  - Each good should be produced by the individual that has the smaller opportunity cost of producing that good

Specialize according to comparative advantage



# Comparative Advantage

- The opportunity cost of a computer is
  - 10 tons of wheat in the U.S.:
    - Producing one computer requires 100 labor hours, which instead could produce 10 tons of wheat
  - 5 tons of wheat in Japan:
    - Producing one computer requires 125 labor hours, which instead could produce 5 tons of wheat

Japan has comparative advantage in computers

美國有50,000 hrs 勞動力
生產一部電腦需要100hrs
生產一噸小麥需要10hrs

日本有30,000 hrs 勞動力
生產一部電腦需要125hrs
生產一噸小麥需要25hrs



# 補充

- 一個國家可以所有產品都有絕對優勢
- 一個國家無法所有產品都有比較優勢
- 一個產品的機會成本是另一個產品其機會成本的倒數



# Comparative Advantage and Trade

總結

- Gains from trade
  - Arise from comparative advantage (differences in opportunity costs)
- When each country specializes in the good(s) in which it has a comparative advantage
  - Total production in all countries is higher
  - The world's "economic pie" is bigger
  - All countries can gain from trade

# Active Learning 4 Absolute and comparative advantage

Argentina, 10,000 hours of labor/month:

- producing 1 lb. coffee requires 2 hours;
- producing 1 bottle wine requires 4 hours

Brazil, 10,000 hours of labor/month:

- producing 1 lb. coffee requires 1 hour
- producing 1 bottle wine requires 5 hours

1. Which country has an absolute advantage in the production of coffee?
2. Which country has a comparative advantage in the production of wine?

# Active Learning 4 Answers

1. Brazil: absolute advantage in coffee
   - Producing 1 lb. coffee:
     - One labor-hour in Brazil, but two in Argentina.
2. Argentina: comparative advantage in wine
   - Argentina's opportunity cost of wine is 2 lb. coffee
     - The four labor-hours required to produce a bottle of wine could instead produce 2 lb. coffee
   - Brazil's opportunity cost of wine is 5 lb. coffee

# Question

- American and Japanese workers can each produce 4 cars a year.
- An American worker can produce 10 tons of grain a year, whereas a Japanese workers can produce 5 tons of grain a year.
- Each country has 100 million workers
  - Which country has an absolute advantage in producing cars? In producing grain?
  - Which country has a comparative advantage in producing cars? In producing grain?

# Question



- Pat and Kris are roommates
- Pat takes 4 hours to brew a gallon of root beer and 2 hours to make a pizza.
- Kris takes 6 hours to brew a gallon of root beer and 4 hours to make a pizza
- The price of pizza can be expressed in terms of gallons of root beer.
- What is the highest price at which pizza can be traded that would make both roommates better off? What is the lowest price?

# Question



- England and Scotland both produce scones and sweaters.
- Suppose that an English worker can produce 50 scones per hour or 1 sweater per hour.
- Suppose that a Scottish worker can produce 40 scones per hour or 2 sweaters per hour.
- If England and Scotland decide to trade, which commodity will Scotland export to England?
- If a Scottish worker could produce only 1 sweater per hour, would Scotland still gain from trade? Would England still gain from trade?

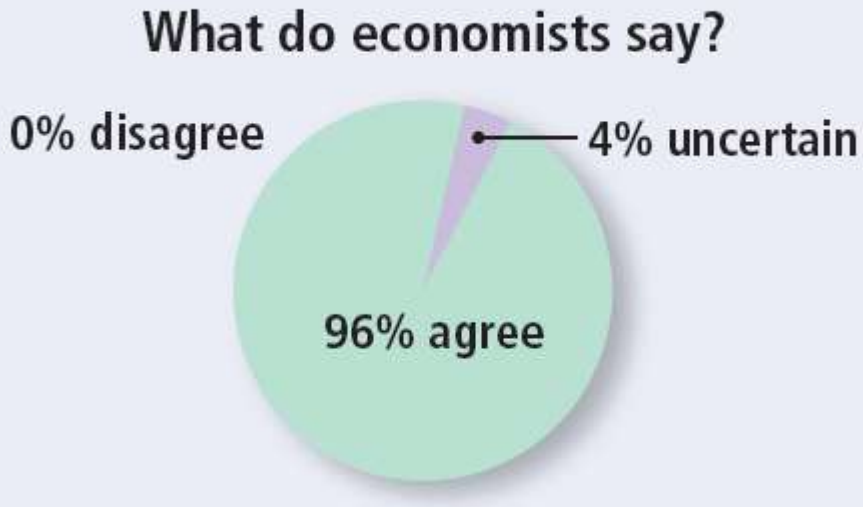
# Question

- A German worker takes 400 hours to produce a car and 2 hours to produce a case of wine.
- A French worker takes 600 hours to produce a car and X hours to produce a case of wine.
- For what values of X will gain from trade be possible?
- For what values of X will Germany export cars and import wine?

# ASK THE EXPERTS

## Trade between China and the United States

*"Some Americans who work in the production of competing goods, such as clothing and furniture, are made worse off by trade with China."*



# Summary

- Interdependence and trade are desirable
  - Allow everyone to enjoy a greater quantity and variety of goods and services
- Comparative advantage: being able to produce a good at a lower opportunity cost
- Absolute advantage: being able to produce a good with fewer inputs (原物料,労力等)
- The gains from trade are based on comparative advantage, not absolute advantage

# Summary

- Trade makes everyone better off
  - It allows people to specialize in those activities in which they have a comparative advantage
- The principle of comparative advantage applies to countries as well as to people
- Economists use the principle of comparative advantage to advocate **free trade** among countries

# Question

- Are the following statements true or false?
  - Two countries can achieve gains from trade even if one of the countries has absolute advantage in the production of all goods
  - Certain talented people have a comparative advantage in everything they do
  - If a certain trade is good for one person, it can't be good for the other one
  - If a certain trade is good for one person, it is always good for the other one
  - If trade is good for a country, it must be good for everyone in the country